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Analysis and Optimization of Influential Factors of Cloud Computing Resource Allocation Based on Random Forests

Fangyuan Cheng
Tianjin University of Technology
Tianjin, China
cfy168@163.com

***Abstract*—With the continuous development of cloud computing technology, efficient resource allocation has become the key to improve cloud computing performance and user satisfaction. In this paper, we deeply analyze the influencing factors of cloud computing resource allocation based on random forest algorithm and propose corresponding optimization strategies. By studying various factors such as resource demand, task type and user behavior in the cloud computing environment, a random forest model is constructed to predict the resource allocation demand in order to achieve more accurate resource allocation. The experimental results show that the method can significantly improve the efficiency and fairness of cloud computing resource allocation, providing strong support for the sustainable development of cloud computing.**

Keywords—Cloud computing, Resource allocation, Random forest, Influencing factors, Optimization strategy

I. Introduction

Cloud computing, as an emerging computing model, has been widely used in various fields. It provides users with convenient, efficient and reliable services by virtualizing and centrally managing computing resources, storage resources and network resources. However, with the increasing number of cloud computing users and the increasingly diversified application requirements, how to effectively allocate resources has become an important challenge for cloud computing. Unreasonable resource allocation not only leads to resource waste and performance degradation, but also affects user experience and satisfaction. Therefore, it is of great practical significance to deeply analyze the influencing factors of cloud computing resource allocation and propose effective optimization strategies to improve the efficiency and quality of cloud computing.

Random forest algorithm, as a powerful machine learning algorithm with the advantages of high accuracy, robustness and easy parallelization, has been widely used in the field of data analysis and prediction. In this paper, the random forest algorithm is applied to cloud computing resource allocation, which provides decision support for realizing more accurate resource allocation by analyzing and predicting the factors affecting resource allocation.

II. Cloud Computing Resource Allocation Overview

A. Concepts of Cloud Computing Resource Allocation

Cloud computing resource allocation refers to the allocation of computing resources, storage resources, network resources, etc. of a cloud computing center to different users and applications in accordance with certain strategies to meet the needs of users and improve the utilization of resources. The goal of resource allocation is to achieve efficient utilization and fair distribution of resources under the premise of ensuring service quality.

B. Characteristics of Cloud Computing Resource Allocation

There are three main methods, namely static allocation, dynamic allocation and hybrid allocation. Static allocation: resources are allocated to users and applications according to a predefined policy at the startup of the cloud computing system. This method is simple and easy to implement, but lacks flexibility and cannot adapt to dynamically changing resource demands. Dynamic allocation: Dynamically adjust the resource allocation strategy based on real-time resource demand and supply. This method can improve resource utilization and service quality, but requires higher computational cost and management complexity. Hybrid allocation: Combining the advantages of static and dynamic allocation, a hybrid allocation is used for resource allocation. For example, static allocation can be performed at system startup and then dynamically adjusted according to the actual situation[1].

C. Factors affecting resource allocation in cloud computing

In the cloud computing resource allocation problem, there are many factors affecting the resource allocation performance, including data resources, performance data, and user demand data. In order to construct a random forest model, we need to extract features from these factors that have a significant impact on resource allocation performance[2]. The specific features are shown in Table I.

TABLE I. Factors affecting resource allocation in cloud computing

Data category	*Data item*	*Descriptive*
Resource data	Computing resources	CPU utilization
	Memory resources	Memory utilization
	Storage resources	Evaluate the effective utilization of storage resources
Performance data	Task response time	Storage resource requirements
	Throughput	Number of tasks processed by the system per unit time
	Error rate	Percentage of errors that occur during task execution
Load data	Server load	Indicates the current working status of the server, including CPU load, memory load, etc.

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Data category	*Data item*	*Descriptive*
	Network load	Reflects the busyness of network transmission, including the amount of data transmission, network delay, etc.
User demand data	User request quantity	The number of tasks submitted by users within a certain period of time.
	User type	For example, ordinary users, VIP users, etc., the demand and priority of resources may be different.
	Changes in user	User demand that changes over time, such as business growth, new service launch, etc.
Resource allocation policy data	Resource partitioning of virtual machines	Resource allocation of virtual machines in terms of CPU, memory, storage, etc.
	Load balancing algorithms	Such as polling algorithm, weighted polling algorithm, minimum number of connections algorithm, etc.
	Resource scheduling policy	Resource scheduling policy designed according to application requirements and priorities.
Cost and benefit data	Resource cost	Including hardware cost, operation and maintenance cost, etc.
	Resource benefits	Business growth, customer satisfaction improvement, etc. through resource allocation
Security and privacy protection data	Resource segregation degree	Resource segregation between different users
	Permission management data	User's permission management for accessing resources
	Encrypted data transmission	Encrypted protection of resources during transmission

III. Random Forest Algorithm

A. Basic Concepts of Random Forests

Random Forest is an integrated learning algorithm that consists of multiple decision trees. Each decision tree is constructed by random sampling and feature selection on the training data. When predicting, Random Forest obtains the final prediction by voting or averaging the predictions of multiple decision trees. In machine learning algorithms, because Random Forest can not only deal with complex, nonlinear and high dimensional problems, but also has a strong generalization ability and robustness[3]. Compared to other algorithms, Random Forest can better deal with overfitting problems and is more tolerant to a large number of input variables and data noise. Therefore, the random forest algorithm is selected to analyze the influence of machine influencing factors on cloud computing resource allocation, in order to reveal the relationship of influencing factors in cloud computing resource allocation more accurately.

B. Random Forest Modeling

Random Forest (RF), as an integrated learning method, has demonstrated excellent prediction and classification capabilities in many domains. By constructing multiple decision trees and synthesizing their results, Random Forest is able to capture nonlinear relationships and complex patterns in data. Its formulation results in:

$$\overline{\mathrm{h}} = \frac{1}{T}\sum_{t=1}^{T}\{h(x,\theta_t)\} \tag{1}$$

- Data Preprocessing: the data of this model is from the API interface of cloud computing platform, log files, and third-party research organizations. The dependent variables are cloud computing resource allocation utilization as well as allocation efficiency, and 8 characteristic variables are screened, which are computing resources, memory resources, task response time, user requests, user demand changes, virtual machine resource division, resource cost and data encrypted transmission. The dataset is divided with the first 80% as the training set and the second 20% as the testing set[4].
- Create the model: to create a random forest model, use the TreeBagger function in MATLAB to create a random forest model, specifying the number of trees and the data set. After the random forest model is created, the predict command can be used to predict the new data and the oobError command can be used to calculate the out-of-bag error, the flowchart is shown in Figure 1.

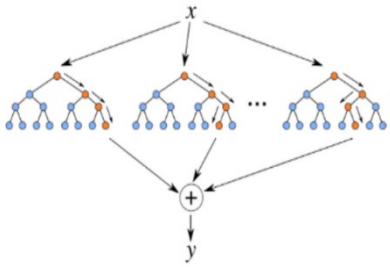

Fig. 1. Schematic diagram of random forest

- Analysis of Importance Calculation: the feature importance of Random Forest is calculated by the built-in OOB Permuted Pr edictor DeltaError property. This property calculates the Permutation Feature Importance.
- The OOB Permuted Pr edictor DeltaError is calculated by the following steps: first for each feature, its value in the OOB sample is randomly perturbed (i.e., displacing the feature value), and the other features are kept unchanged. The difference between the model's prediction error on the OOB sample and the original prediction error is then computed after upsetting a feature value. This difference measures how much the model's performance decreases after upsetting a feature value, and can therefore be considered as the importance of that feature. Finally, the above steps are repeated to displace all features to obtain the importance of each feature.
- Specific steps: Original OOB Error (Original OOB Error) calculation; first train the Random Forest model using all the data and labels, and set oob_ score = Ture. equation 1 calculates the Original OOB Error OOBoriginal [5], which is usually used as the Mean Square Error (MSE).Where y_i is the true label and $\hat{y}_{oob,t}$ is the OOB prediction.

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$$OBB_{original}=\frac{1}{N}\sum_{i=1}^{N}(y_i-\hat{y}_{OOb,t})^2 \quad (2)$$

- Secondly, the Permutation Feature Importance (PFI) is calculated: in equation (2) for each feature f i.e., the original feature data is copied. In the copied data, the value of feature f is randomly disrupted. Retrain the model using the data with the disrupted feature values. Compute the new OOB error , $OOB_{permuted,f}$. Compute the Delta Error $\Delta Error_f$ .Where the $\Delta Error_f$ value indicates the importance of feature *f*; the larger the value, the more important the feature.

$$\Delta Error_f=OOB_{permuted,f}-OBB_{original} \quad (3)$$

C. Random Forest Model Importance Analysis

The Random Forest model constructed in the previous paper is simulated, and the data set is divided, with the first 80% as the training set and the second 20% as the test set, and the R2 of the data in the training set is calculated to be 0.9522; the R2 of the data in the test set is calculated to be 0.86157; the MAE of the data in the training set is 0.02025; the MAE of the data in the test set is 0.031289; and the RMSE of the data in the training set is calculated to be 0.038698; the RMSE of the data in the test set is 0.025317. 0.038698 for the training set data and 0.025317 for the test set data, and the predictions are as expected. Finally the importance calculation is able to represent the histogram of the importance of the feature parameters, as shown in Figure 2.

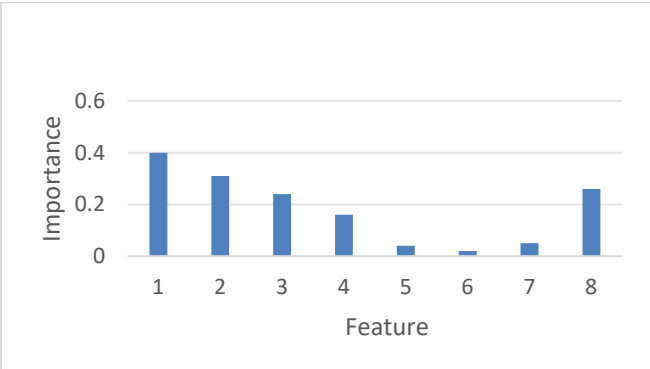

Fig. 2. Influential histogram of characteristic parameters of cloud computing resource utilization efficiency

Characteristic parameters 1-8 in the figure are represented as computing resources, memory resources, task response time, user request volume, user demand change, virtual machine resource division, resource cost and data encrypted transmission situation respectively. From Fig. 2, it can be seen that the characteristic parameters of 1-4 and 8 have a greater impact on resource utilization, especially 1 and 2. Therefore, optimizing computing resources, memory resources and data encrypted transmission can effectively improve resource utilization and effectively optimize cloud computing resource allocation[6]. The same model construction and parameter optimization methods are used to analyze the influencing factors of resource allocation rate and calculate the importance degree.

The data set is also divided, with the first 80% as the training set and the second 20% as the test set. The results of the resource allocation efficiency test set and the results of the resource allocation efficiency training set were obtained, and the R 2 of the training set data was calculated as 0.853; the R 2 of the test set data was calculated as 0.8521; the MAE of the training set data was 0.0213; the MAE of the test set data was 0.0323; the RMSE of the training set data was 0.0376; and the RMSE of the test set data was 0.0262. The predictions are in line with expectations. Finally the importance calculation is able to represent the histogram of the importance of the feature parameters, as shown in Figure 3.

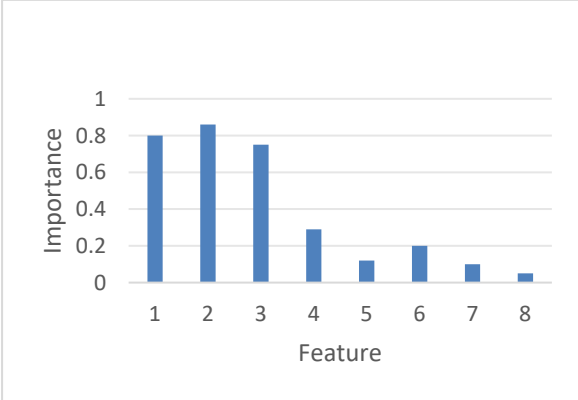

Fig. 3. Influence histogram of parameters characterizing resource allocation efficiency in cloud computing

Still the characteristic parameters in the figure are represented as computational resources, memory resources, task response time, user requests, user demand variation, virtual machine resource partitioning, resource cost and data encrypted transmission. From the figure, it can be seen that memory resources have a greater impact on resource allocation efficiency, followed by computational resources and task response time. So optimizing computing resources, memory resources and task response time can effectively improve resource allocation efficiency and effectively affect cloud computing resource allocation.

IV. Random Forest Based Resource Allocation Optimization

After analyzing the influencing factors of cloud computing resource allocation, the resource allocation strategy is optimized using the random forest model. The goal of the optimization is to maximize the utilization rate and allocation efficiency of the resources under the premise of satisfying the users' needs.

A. Optimization strategy

The resource allocation optimization strategy based on the random forest model mainly includes the following aspects: first, resource selection optimization: according to the results of the feature importance assessment, priority is given to allocating resources that have a greater impact on resource allocation performance[7]. For example, in the case of resource tension, priority can be given to allocating resources with higher contribution to resource utilization and allocation efficiency. Second, task scheduling optimization: according to the resource demand and resource attribute characteristics of the task, a reasonable task scheduling strategy is formulated. For example, tasks with similar resource requirements can be merged to reduce resource wastage and conflicts; at the same time, tasks can also be sorted according to priority to ensure that important tasks can obtain resources on a priority basis. Third, parameter tuning: the parameters of the resource allocation strategy are

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optimized using the random forest model. For example, parameters such as the number and maximum depth of decision trees can be adjusted to improve the accuracy and stability of the model; at the same time, the selection of allocation algorithms and parameter settings can also be adjusted to optimize the performance of resource allocation[8].

B. Experimental design and realization

In order to verify the effectiveness of the resource allocation optimization method based on random forest, the following experiments are designed.

1) Dataset Preparation: collect resource usage data and task requirement data on the cloud computing platform as a dataset for the experiment. The dataset should include different types of resources (e.g., CPU, memory, storage, etc.) and tasks (e.g., compute-intensive tasks, IO-intensive tasks, etc.).

2) Model Training and Evaluation: the dataset is utilized to construct a random forest model, and model training and evaluation is performed. During the training process, we can utilize methods such as cross-validation to evaluate the performance of the model; at the same time, we can also utilize the results of feature importance assessment to identify the main features that affect the performance of resource allocation.

3) Optimization strategy implementation: According to the feature importance assessment results and optimization strategy, the resource allocation strategy can be adjusted and optimized. For example, priority can be given to allocating resources that contribute more to resource utilization and allocation efficiency; at the same time, task scheduling and parameter settings can also be adjusted to optimize the performance of resource allocation.

C. Experimental results and analysis

1) Resource Utilization Improvement: the formula of resource utilization is: (Improved Resource Utilization - Initial Resource Utilization/Initial Resource Utilization) × 100%, and Table II and III shows that after optimization, the resource utilization of the cloud computing platform improves as: (0.72-0.6/0.6) × 100% = 20%. This shows that the optimization strategy can effectively improve the utilization efficiency of resources and reduce the waste and conflict of resources.

2) Improved resource allocation efficiency: resource allocation efficiency is measured by utilizing the total task completion time (T). The optimized resource allocation strategy also shows significant improvement in terms of allocation efficiency. As shown in Table II and III, the percentage reduction in task completion time compared to pre-optimization is: (2-1.7)/2 × 100% = 15%. This indicates that the optimization strategy can speed up the allocation and execution of tasks and improve the responsiveness and service quality of the system, thus effectively improving the resource allocation efficiency.

TABLE II. PRE-OPTIMIZATION RESOURCE DATA

Resource Type Data Item Numeric	***Data item***	***Descriptive***
CPU resources	Total input	1000 cores
	Utilization rate	60%
Memory resources	Total input	20TB
	Utilization rate	50%
Storage space	Total input	100PB
	Utilization rate	40%
Business Transaction Volume	Transactions per second	10000 transactions per second
Service quality	Response time	Average 0.1 seconds
	Error rate 0.01%	
Task completion	Number of tasks	100
	Average completion time per task	2h
	Total task completion time	200h

TABLE III. OPTIMIZED RESOURCE DATA

Resource Type Data Item Numeric	***Data item***	***Descriptive***
CPU resources	Total input	1000 cores
	Utilization rate	72%
Memory resources	Total input	20TB
	Utilization rate	60%
Storage space	Total input	100PB
	Utilization rate	50%
Business Transaction Volume	Transactions per second	12000 transactions per second
Service quality	Response time	Average 0.1 seconds
	Error rate 0.01%	
Task completion	Number of tasks	100
	Average completion time per task	2h
	Total task completion time	200h

V. CONCLUSIONS AND OUTLOOK

A. Research summary

This paper deeply analyzes the influencing factors of cloud computing resource allocation based on random forest and proposes corresponding optimization strategies. By studying various factors such as resource demand, task type and user behavior in the cloud computing environment, a random forest model is constructed to predict the resource allocation demand in order to achieve more accurate resource allocation. The experimental results show that the method can significantly improve the efficiency and fairness of cloud computing resource allocation, providing strong support for the sustainable development of cloud computing.

B. Research Outlook

1) The random forest model is further optimized to improve the performance and generalization ability of the model. More advanced feature selection algorithms, model optimization algorithms, etc. can be used to improve the

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accuracy and efficiency of the model.

2) To combine other machine learning algorithms and optimization algorithms to build a more intelligent cloud computing resource allocation system. Deep learning algorithms, reinforcement learning algorithms, etc. can be combined with random forest algorithms to improve the intelligence of resource allocation.

3) More comprehensive cloud computing resource allocation model is constructed by considering more influencing factors, such as energy consumption, cost-effectiveness and so on. Factors such as energy consumption and cost-effectiveness can be included in the consideration of resource allocation to realize the efficient use of resources and sustainable development.

4) Research on the dynamics and real-time nature of cloud computing resource allocation should be strengthened to improve the responsiveness and adaptability of resource allocation. Real-time monitoring and forecasting techniques can be adopted to achieve dynamic adjustment and optimization of resource allocation.

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